4.2 Cockroach - *Periplaneta americana*

Classification		
Phylum	:	Arthropoda
Class	:	Insecta
Order	:	Orthoptera
Genus	:	*Periplaneta*
Species	:	*americana*

Cockroach is a typical cosmopolitan insect and exhibits all the fundamental characteristics of Class Insecta. Generally cockroaches are reddish brown or black bodied with a light brown margin in the first thoracic segment. They are omnivores, nocturnal, living in damp and warm places and are quite common in kitchens, hotels, bakeries, restaurants, warehouse, sewage and public places. *Periplaneta* is a cursorial (swift runner) animal. It is dioecious and oviparous and exhibits parental care. They carry with them harmful germs of various bacterial diseases like cholera, diarrhoea, tuberculosis, and typhoid and hence are known as "**Vectors**".

Morphology

The adult cockroaches are about 2 to 4 cm in length and about 1cm in width. The body of the cockroach is compressed dorso-ventrally, bilaterally symmetrical, segmented and is divisible into three distinct regions – head, thorax and abdomen. The entire body is covered by a hard, brown coloured, chitinous exoskeleton. In each segment, exoskeleton has hardened plates called **sclerites**, which are joined together by a delicate and elastic **articular membrane** or **arthrodial membrane**. The sclerites of the dorsal side are called **tergites**, those on the ventral side are called **sternites** and those of lateral sides are called **pleurites**.

The head of cockroach is small, triangular lies at right angle to the longitudinal body **axis**. the mouth parts are directed downwards so it is **hypognathous.** It is formed by the fusion of six segments and shows great mobility in all directions due to a flexible neck (Figure 4.8). The head capsule bears a pair of large, sessile, and reniform **compound eyes**, a pair of

The cockroaches are ancient and most basic among all groups of insects, dating back to the carboniferous period, about 320 million years ago.

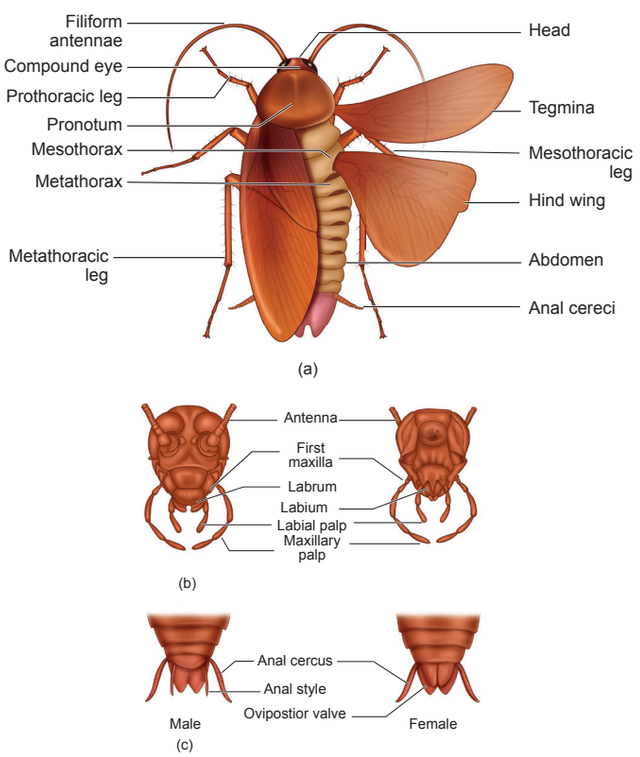


Figure 4.8 *Periplaneta americana:* (a) External features (b) Head dorsal and ventral view (c) Male and Female ventral view of posterior segment of abdomen

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